

```

# Importing Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Realistic IPL Sample Data
data = {
    'season': [2023, 2023, 2022, 2022, 2021, 2021, 2020, 2020, 2019],
    'team1': ['MI', 'RCB', 'CSK', 'DC', 'KKR', 'SRH', 'RR', 'PBKS', 'GT'],
    'team2': ['CSK', 'MI', 'RCB', 'MI', 'DC', 'RCB', 'MI', 'KKR', 'LSG'],
    'winner': ['CSK', 'RCB', 'CSK', 'MI', 'DC', 'RCB', 'MI', 'KKR', 'GT'],
    'city': ['Mumbai', 'Bangalore', 'Chennai', 'Delhi', 'Kolkata', 'Hyderabad', 'Jaipur', ''],
    'player_of_match': ['Ruturaj Gaikwad', 'Faf du Plessis', 'Moeen Ali', 'Suryakumar Yadav',
                        'Glenn Maxwell', 'Jasprit Bumrah', 'Andre Russell', 'Shubman Gill'],
    'runs_scored': [218, 190, 205, 198, 210, 187, 199, 202, 215],
}

# Create DataFrame
ipl_df = pd.DataFrame(data)

# 1. Basic Exploration
print("Data Info:")
ipl_df.info()
print("\nFirst 5 Rows:")
print(ipl_df.head())

# 2. Team Performance
team_wins = ipl_df['winner'].value_counts()
print("\nTeam Wins:\n", team_wins)

# 3. Player of the Match Counts
player_of_match_counts = ipl_df['player_of_match'].value_counts()
print("\nPlayer of the Match Counts:\n", player_of_match_counts)

# 4. Average Runs Analysis
average_runs_per_match = ipl_df['runs_scored'].mean()
print(f"\nAverage Runs Scored per Match: {average_runs_per_match:.1f}")

# 5. Bar Chart of Team Wins
plt.figure(figsize=(10, 6))
sns.countplot(x='winner', data=ipl_df, order=team_wins.index, palette='viridis')
plt.title('Number of Wins by Team')
plt.xlabel('Team')
plt.ylabel('Number of Wins')
plt.show()

# 6. Histogram of Runs
plt.figure(figsize=(8, 5))
plt.hist(ipl_df['runs_scored'], bins=6, color='skyblue', edgecolor='black')
plt.title('Distribution of Runs Scored per Match')
plt.xlabel('Runs Scored')
plt.ylabel('Frequency')
plt.show()

```



Data Info:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 9 entries, 0 to 8

Data columns (total 7 columns):

#	Column	Non-Null Count	Dtype
0	season	9 non-null	int64
1	team1	9 non-null	object
2	team2	9 non-null	object
3	winner	9 non-null	object
4	city	9 non-null	object
5	player_of_match	9 non-null	object
6	runs_scored	9 non-null	int64

dtypes: int64(2), object(5)

memory usage: 636.0+ bytes

First 5 Rows:

	season	team1	team2	winner	city	player_of_match	runs_scored
0	2023	MI	CSK	CSK	Mumbai	Ruturaj Gaikwad	218
1	2023	RCB	MI	RCB	Bangalore	Faf du Plessis	190
2	2022	CSK	RCB	CSK	Chennai	Moeen Ali	205
3	2022	DC	MI	MI	Delhi	Suryakumar Yadav	198
4	2021	KKR	DC	DC	Kolkata	Prithvi Shaw	210

Team Wins:

winner

CSK 2

RCB 2

MI 2

DC 1

KKR 1

GT 1

Name: count, dtype: int64

Player of the Match Counts:

player_of_match

Ruturaj Gaikwad 1

Faf du Plessis 1

Moeen Ali 1

Suryakumar Yadav 1

Prithvi Shaw 1

Glenn Maxwell 1

Jasprit Bumrah 1

Andre Russell 1

Shubman Gill 1

Name: count, dtype: int64

Average Runs Scored per Match: 202.7

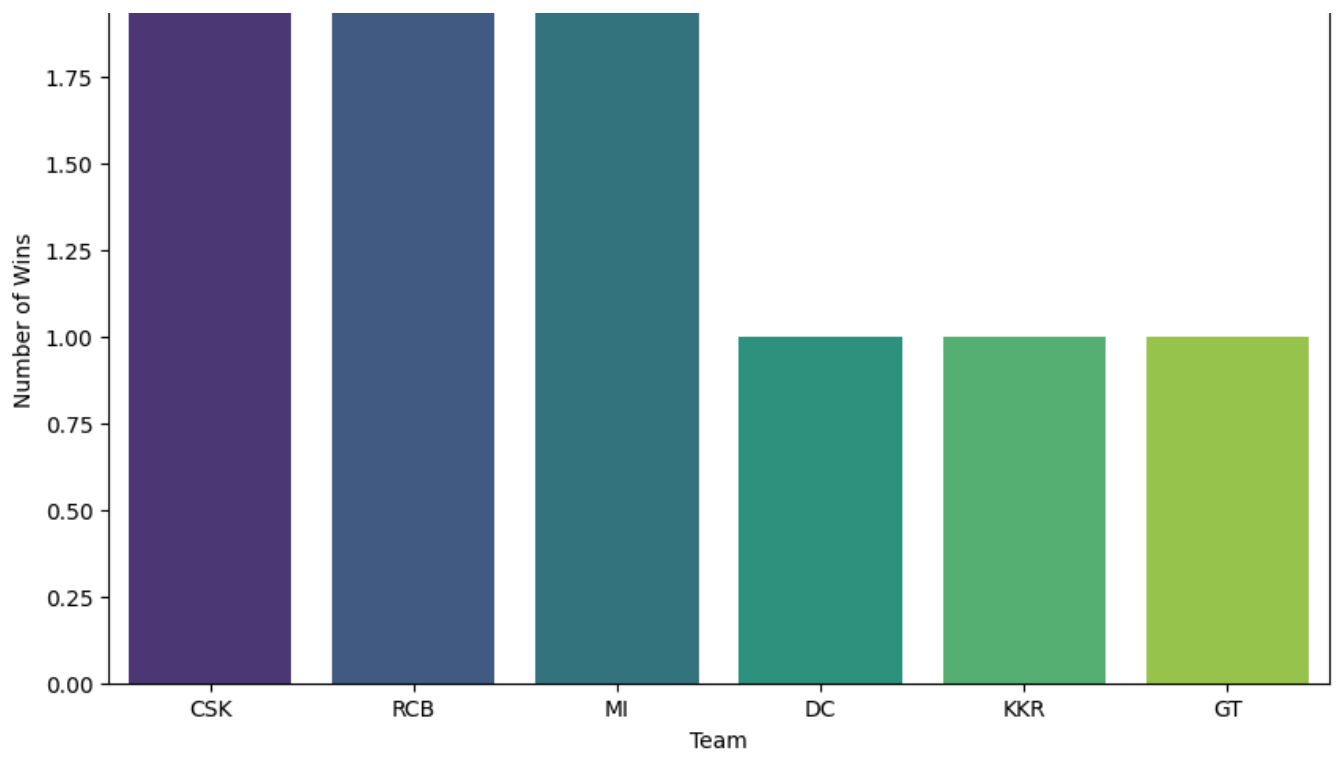
<ipython-input-2-9dc11582664a>:41: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v

sns.countplot(x='winner', data=ipl_df, order=team_wins.index, palette='viridis

Number of Wins by Team





Distribution of Runs Scored per Match